

Crystal Structure

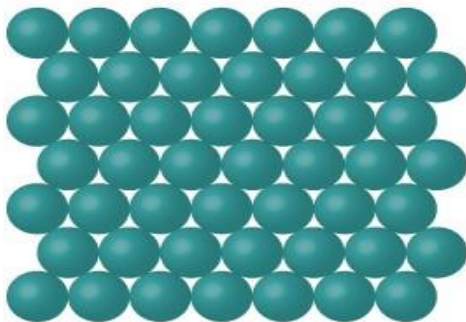
3.1 Fundamental Concepts:

Solid materials may be classified according to the regularity with which atoms or ions are arranged with respect to one another into two groups:

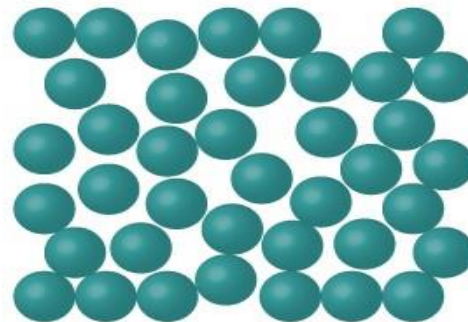
- 1- Crystalline materials
- 2- Non-Crystalline materials

1-Crystalline Material: is one in which the atoms are situated in a repeating or periodic array over large atomic distances; that is, long-range order exists, such that upon solidification, the atoms will position themselves in a repetitive three-dimensional pattern, in which each atom is bonded to its nearest-neighbor atoms. A crystalline material may be either in the form of single crystal or an aggregate of many crystals known as polycrystalline. The polycrystalline material is always separated by well-defined boundaries.

Non –Crystalline materials: lack a systematic and regular arrangement of atoms over relatively large atomic distance (short range order (SRO) ,such as plastic, glass, etc.



Crystalline

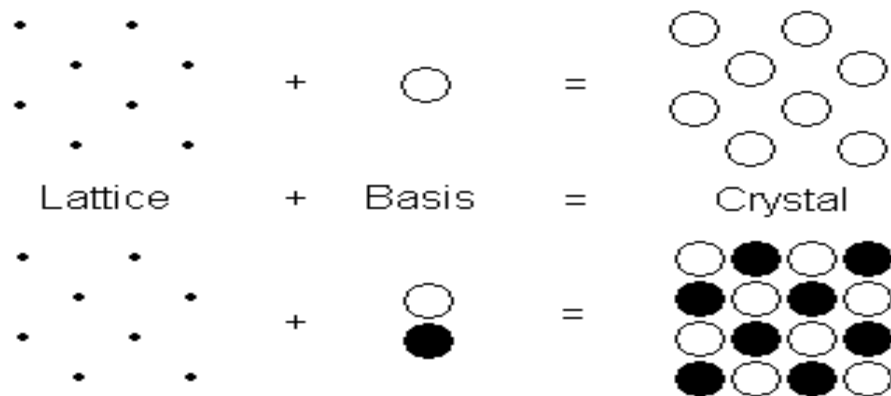


Amorphous

Crystal structure: the manner in which atoms, ions, or molecules are spatially arranged. There are an extremely large number of different crystal structures all having long- range atomic order or the crystal structure is described in terms of a lattice with a group of atomic attached to each lattice point. The group is called the basis, it is repeated in space to form the crystal structure.

Lattice: is used in the context of crystal structures; in this sense “lattice” means a three-dimensional array of points coinciding with atom positions (or sphere centers).is pattern (or net)of points having a geometrical concepts, it has no spherical meaning, each point in the lattice is called lattice point.

Basis: is an assembly (group) of atoms or molecules identical in composition arrangement and in their orientation.



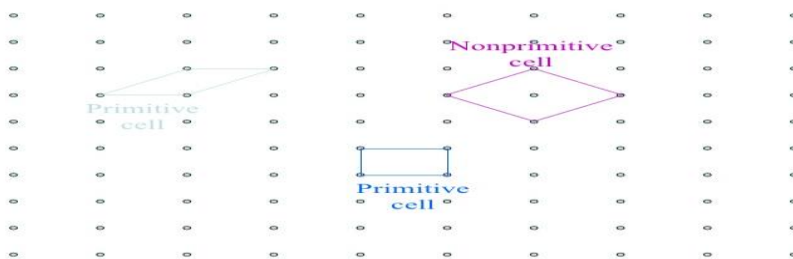
Unit cell: small groups of atoms form a repetitive pattern, thus, the unit cell is the basic structural unit or building block of the crystal structure, it is chosen to represent the symmetry of the crystal structure and there are two kind of unit cell:

1. Primitive cell:

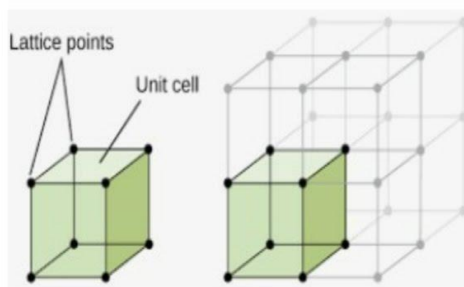
Is the smallest unit cell in the crystal, usually there are four points at the corners in (2-D) (possess lattice points at its corners only), it is the simplest type of unit cell which contain one lattice point per unit cell. There are 8 points at the corners in (3-D) also one point per unit cell (each corner point is shared among the eight cell).

2. Non-Primitive cell:

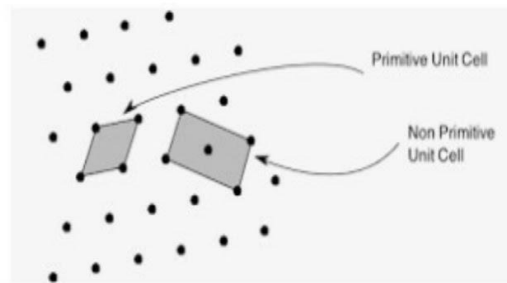
It is the larger unit cell than the primitive cell; there are additional points to that at the corners. The area of the non-primitive cell must be an integral multiple of primitive cell. **Prove it**



Unit Cell



Primitive Unit Cell, Non Primitive Unit Cell



3.2 Types of Crystal Systems:

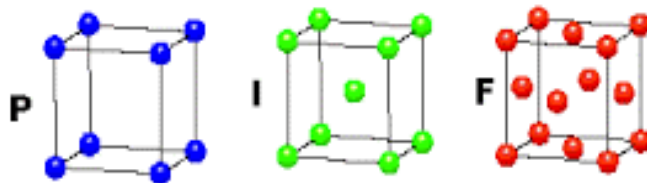
In 3-D , there are 14 Bravais lattice types are grouped into 7 systems according to the 7 types of the conventional unit cells , Triclinic , Monoclinic, orthorhombic ,Tetragonal , cubic, Trigonal and Hexagonal.

The magnitude a,b,c are called the lattice parameters, α, β, γ are called interfacial angles.

CUBIC

$$a = b = c$$

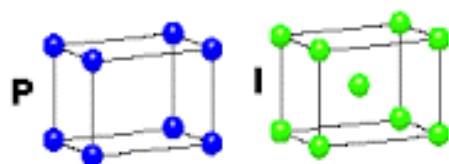
$$\alpha = \beta = \gamma = 90^\circ$$



TETRAGONAL

$$a = b \neq c$$

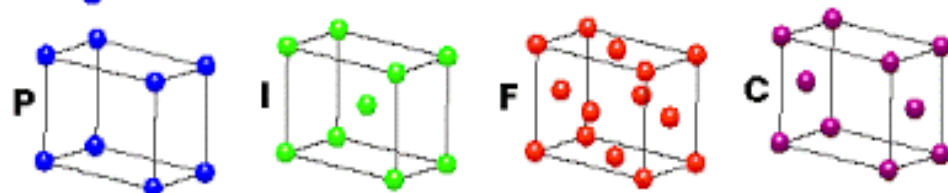
$$\alpha = \beta = \gamma = 90^\circ$$



ORTHORHOMBIC

$$a \neq b \neq c$$

$$\alpha = \beta = \gamma = 90^\circ$$

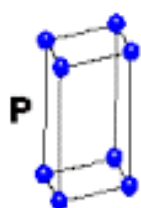


HEXAGONAL

$$a = b \neq c$$

$$\alpha = \beta = 90^\circ$$

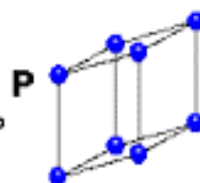
$$\gamma = 120^\circ$$



TRIGONAL

$$a = b = c$$

$$\alpha = \beta = \gamma \neq 90^\circ$$

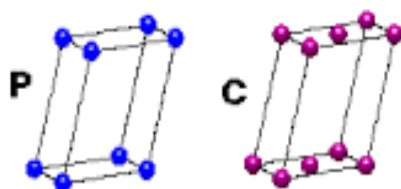


MONOCLINIC

$$a \neq b \neq c$$

$$\alpha = \gamma = 90^\circ$$

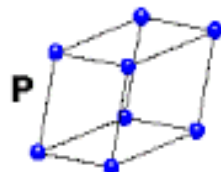
$$\beta \neq 120^\circ$$



TRICLINIC

$$a \neq b \neq c$$

$$\alpha \neq \beta \neq \gamma \neq 90^\circ$$



4 Types of Unit Cell

P = Primitive

I = Body-Centred

F = Face-Centred

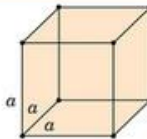
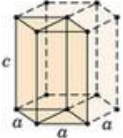
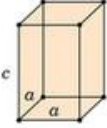
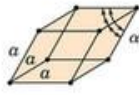
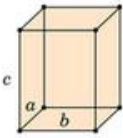
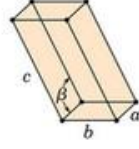
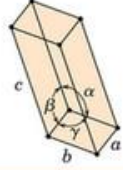
C = Side-Centred

+

7 Crystal Classes

→ 14 Bravais Lattices

Lattice Parameter Relationships and Figures Showing Unit Cell Geometries for the Seven Crystal Systems

<i>Crystal System</i>	<i>Axial Relationships</i>	<i>Interaxial Angles</i>	<i>Unit Cell Geometry</i>
 Cubic	$a = b = c$	$\alpha = \beta = \gamma = 90^\circ$	
 Hexagonal	$a = b \neq c$	$\alpha = \beta = 90^\circ, \gamma = 120^\circ$	
 Tetragonal	$a = b \neq c$	$\alpha = \beta = \gamma = 90^\circ$	
 Rhombohedral (Trigonal)	$a = b = c$	$\alpha = \beta = \gamma \neq 90^\circ$	
 Orthorhombic	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90^\circ$	
 Monoclinic	$a \neq b \neq c$	$\alpha = \gamma = 90^\circ \neq \beta$	
 Triclinic	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90^\circ$	

3.3 Metallic Crystal Structures:

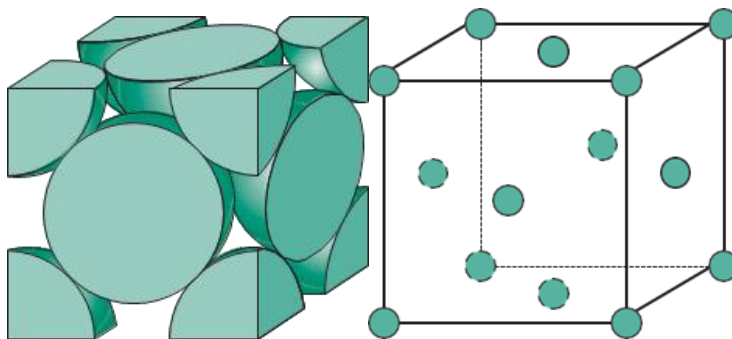
The atomic bonding in this group of materials is metallic and thus no directional in nature. Consequently, there are minimal restrictions as to the number and position of nearest-neighbor atoms; this leads to relatively large numbers of nearest neighbors and dense atomic packings for most metallic crystal structures. Also, for metals, using the hard sphere model for the crystal structure, each sphere represents an ion core. Three relatively simple crystal structures are found for most of the common metals:

- 1- Face centeedr cubic Fcc
- 2- Body centered cubic Bcc
- 3- Hexagonal close-packed Hcp

1-Face-Centered Cubic Crystal Structure (FCC)

The crystal structure found for many metals has a unit cell of cubic geometry, with atoms located at each of the corners and the centers of all the cube faces. It is aptly called the face-centered cubic (FCC) crystal structure. Some of the familiar metals having this crystal structure are copper, aluminum, silver, and gold.

For the FCC crystal structure, each corner atom is shared among eight unit cells, whereas a face-centered atom belongs to only two. Therefore, one-eighth of each of the eight corner atoms and one-half of each of the six face atoms, or a total of four whole atoms, may be assigned to a given unit cell.



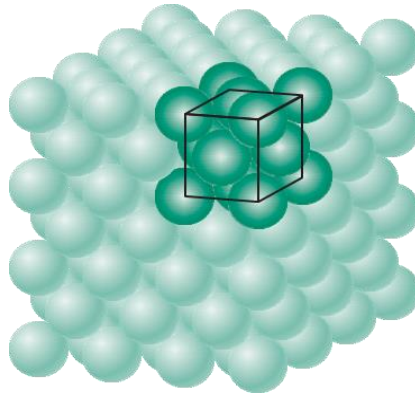
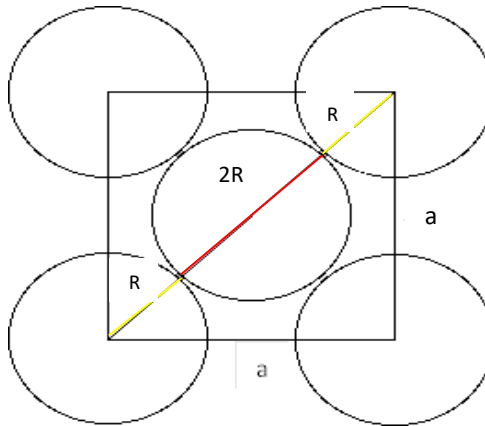


Figure 3.1 face- centered cubic crystal structure, (a) a hard sphere unit cell representation, (b) a reduced-sphere unit cell, and (c) an aggregate of many atoms.



a = length of cubic
 R = radius of atom

1- To find the relation between (a) and (R) applying Pythagoras theory:

$$(R + 2R + R)^2 = a^2 + a^2$$

$$(4R)^2 = 2a^2$$

$$16R^2 = 2a^2 \quad \text{by the square root of both sides yield :} \quad \sqrt{2} a = 4R$$

2- To find number of atoms per unit cell

The atoms of each corner is shared among 8 unit cell, While 6 atoms at faces is belong to 2 unit cell. Thus:

$$\text{Number of atoms per unit cell is} = \left(\frac{1}{8} \times 8\right) + \left(\frac{1}{2} \times 6\right) = 4$$

3- To find Atomic packing factor:

$$\text{APF} = \frac{\text{Volume of atoms in unit cell}}{\text{Total unit cell volume}}$$

$$\text{APF} = \frac{\frac{4}{3}\pi R^3 \times 4}{a^3} = \frac{\frac{4}{3}\pi R^3 \times 4}{16\sqrt{2}R^3} = 0.74$$

2-Body-Centered Cubic Crystal Structure (BCC):

Another common metallic crystal structure also has a cubic unit cell with atoms located at all eight corners and a single atom at the cube center. This is called a body-centered cubic (BCC) crystal structure. A collection of spheres depicting this crystal structure is shown in Figure 3.2c, whereas Figures 3.2a and 3.2b are diagrams of BCC unit cells with the atoms represented by hard sphere.

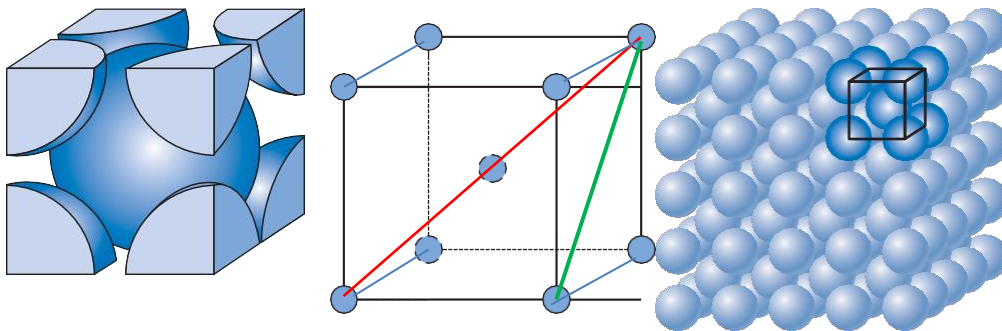
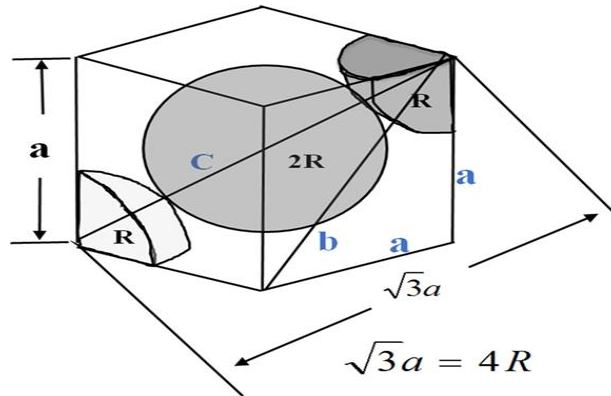


Figure 3.2 For the body-centered cubic crystal structure, (a) a hard sphere unit cell representation, (b) a reduced-sphere unit cell, and (c) an aggregate of many atoms.



a = unit cell length , R = atomic radius , red line = L and green line = H

a- To find the relation between a and R : applying Pythagoras theory twice as:

$$H^2 = a^2 + a^2 \quad \longrightarrow \quad H^2 = 2a^2$$

$$L^2 = a^2 + H^2 \quad \longrightarrow \quad a^2 + 2a^2 \quad \longrightarrow \quad 3a^2 \quad \text{While as } L = 4R \text{ then:}$$

$$(4R)^2 = 3a^2 \quad \longrightarrow \quad a = 4R/\sqrt{3}$$

b- No. of atoms per unit cell

Eight (8) atoms at corners shared to 8 unit cell and one (1) atom in center belong to unit cell thus:

$$\text{No. of } \frac{\text{atoms}}{\text{unit cell}} = \left(8 \times \frac{1}{8} \right) + 1 = 2$$

c- To find Atomic packing factor:

$$APF = \frac{\frac{4}{3}\pi R^3 \times \frac{\text{No. of atoms}}{\text{unit cell}}}{\text{Volume}}$$

$$APF = \frac{\frac{4}{3}\pi R^3 \times 2}{a^3}$$

$$APF = \frac{\frac{4}{3}\pi R^3 \times 2}{\left(\frac{4R}{\sqrt{3}}\right)^3} = 0.68$$

Note: APF for BCC less than APF for FCC, Why?

H.W: Calculate with the same method the relation between (a and R) ,No. of atoms /unit cell and APF for Simple cubic (SC) structure.

3-4 Density Computations:

A knowledge of the crystal structure of a metallic solid permits computation of its theoretical density r through the relationship:

$$r = \frac{nA}{V_c N_a}$$

Where

n = number of atoms associated with each unit cell

A = atomic weight

V_c = Volume of the unit cell

N_a = Avogadro's number = 6.023×10^{23} atoms/mol

H.W: Copper has an atomic radius of 0.128 nm, an FCC crystal structure, and an atomic weight of 63.5 g/mol. Compute its theoretical density.

Important Properties of unit cell

	SC	BCC	FCC	HCP
Relation between atomic radius (r) and lattice parameter (a)	$a = 2r$	$\sqrt{3}a = 4r$	$\sqrt{2}a = 4r$	$a = 2r$
Effective Number of atoms per unit cell	1	2	4	6
Coordination Number (No. of nearest equidistant neighbours)	6	8	12	12
Packing Factor	0.52	0.68	0.74	0.74